

Администрирование локальных сетей

Лабораторная работа № 6. Статическая маршрутизация VLAN

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Настроить статическую маршрутизацию между виртуальными локальными сетями (VLAN) в среде Cisco Packet Tracer с использованием технологии Router on a Stick и провести анализ передачи данных между различными VLAN.

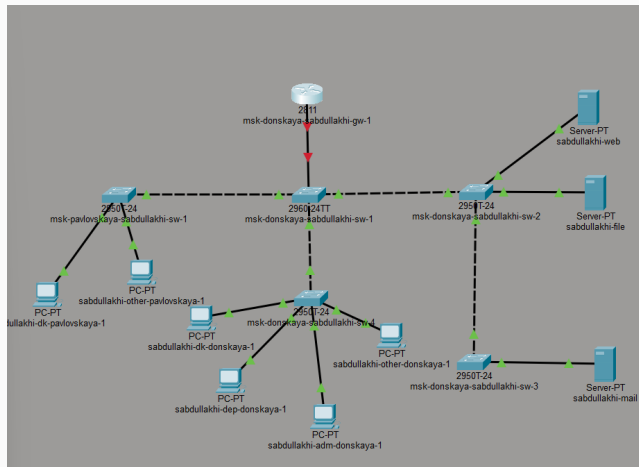


Рис. 1: топологии сети

Первичная настройка маршрутизатора

```
Router>ena
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-sabdullakhi-gw-1
msk-donskaya-sabdullakhi-gw-1(config)#line vty 0 4
msk-donskaya-sabdullakhi-gw-1(config-line)#pass cisco
msk-donskaya-sabdullakhi-gw-1(config-line)#login
msk-donskaya-sabdullakhi-gw-1(config-line)#line console 0
msk-donskaya-sabdullakhi-gw-1(config-line)#pass cisco
msk-donskaya-sabdullakhi-gw-1(config-line)#login
msk-donskaya-sabdullakhi-gw-1(config-line)#ena secret cisco
msk-donskaya-sabdullakhi-gw-1(config)#service password-encryption
^
% Invalid input detected at '^' marker.

msk-donskaya-sabdullakhi-gw-1(config)#service password-encryption
msk-donskaya-sabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-sabdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-sabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-sabdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-sabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:4:32.246: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:4:32.246: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-sabdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-sabdullakhi-gw-1(config-line)#
msk-donskaya-sabdullakhi-gw-1(config-line)#exit
```

Рис. 2: Первичная настройка маршрутизатора через CLI

Первичная настройка маршрутизатора

```
msk-donskaya-sabdullakhi-gw-1(config)#interface f0/0
msk-donskaya-sabdullakhi-gw-1(config-if)#no shutdown

msk-donskaya-sabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up

msk-donskaya-sabdullakhi-gw-1(config-if)#interface f0/0.2
msk-donskaya-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state
to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 2
msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.1.1 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(config-subif)#description management
msk-donskaya-sabdullakhi-gw-1(config-subif)#interface f0/0.3
msk-donskaya-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state
to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 3
msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(config-subif)#description servers
msk-donskaya-sabdullakhi-gw-1(config-subif)#
```

Рис. 3: Первичная настройка маршрутизатора через CLI

Настройка межсетевой маршрутизации VLAN (Router on a Stick)

```
msk-donskaya-sabdullakhi-gw-1(config-subif)#interface f0/0.101
msk-donskaya-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.101, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state
to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 101
msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(config-subif)#description dk
msk-donskaya-sabdullakhi-gw-1(config-subif)#
msk-donskaya-sabdullakhi-gw-1(config-subif)#interface f0/0.102
msk-donskaya-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.102, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state
to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 102
msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(config-subif)#description departments
msk-donskaya-sabdullakhi-gw-1(config-subif)#
msk-donskaya-sabdullakhi-gw-1(config-subif)#interface f0/0.103
msk-donskaya-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.103, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state
to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 103
msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(config-subif)#description adm
msk-donskaya-sabdullakhi-gw-1(config-subif)#
msk-donskaya-sabdullakhi-gw-1(config-subif)#interface f0/0.104
msk-donskaya-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.104, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state
to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 104
msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(config-subif)#description other
msk-donskaya-sabdullakhi-gw-1(config-subif)#
msk-donskaya-sabdullakhi-gw-1(config-subif)#exit
msk-donskaya-sabdullakhi-gw-1(config)#exit
msk-donskaya-sabdullakhi-gw-1#
%SYS-5-CONFIG I: Configured from console by console
```

Проверка доступности устройств (ping)

```
sabdullakhi-other-pavlovskaya-1

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 10.128.4.10

Pinging 10.128.4.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time=7ms TTL=127

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>ping 10.128.5.10

Pinging 10.128.5.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127
Reply from 10.128.5.10: bytes=32 time=1ms TTL=127

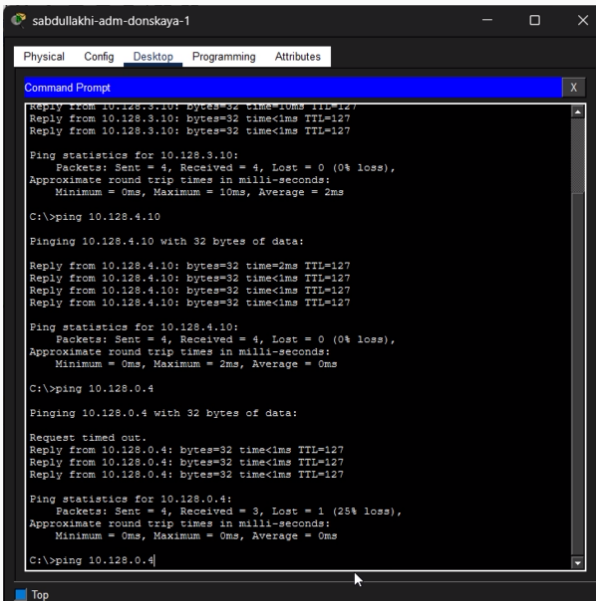
Ping statistics for 10.128.5.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.6.10

Pinging 10.128.6.10 with 32 bytes of data:

Reply from 10.128.6.10: bytes=32 time<1ms TTL=128
Reply from 10.128.6.10: bytes=32 time<1ms TTL=128
Reply from 10.128.6.10: bytes=32 time<1ms TTL=128
Reply from 10.128.6.10: bytes=32 time<1ms TTL=128
```

Дополнительно выполнена проверка



The screenshot shows a Windows Command Prompt window titled 'Command Prompt' with a tab bar at the top containing 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The window displays the results of three ping commands. The first two pings (to 10.128.3.10 and 10.128.4.10) are successful with 0% loss. The third ping (to 10.128.0.4) shows a 25% loss, with one request timing out.

```
sabdullakhi-adm-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 10.128.3.10: bytes=32 time=10ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 10.128.4.10

Pinging 10.128.4.10 with 32 bytes of data:

Reply from 10.128.4.10: bytes=32 time=2ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.4
```

Анализ передачи данных в режиме Simulation

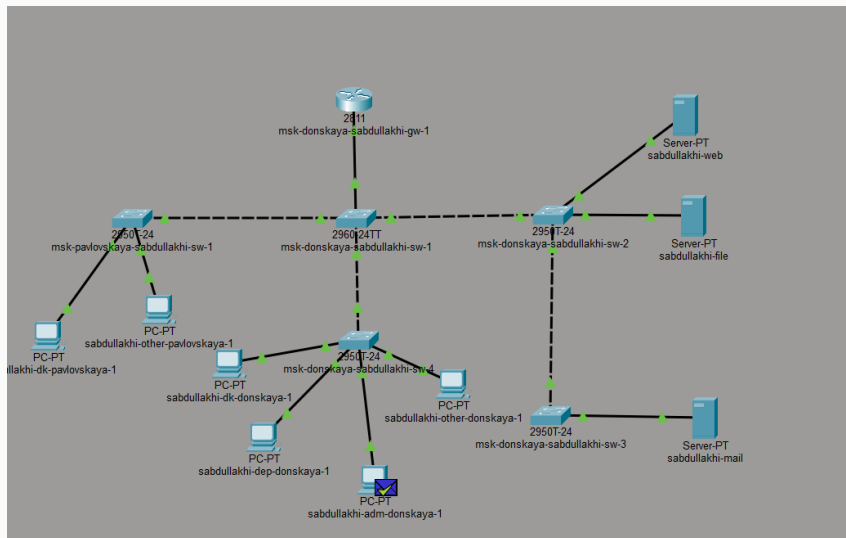


Рис. 7: Просмотр перемещения пакета в режиме Simulation

Анализ передачи данных в режиме Simulation

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	sabdullakhi-adm-donskaya-1	ICM
	0.001	sabdullakhi-adm-donskaya-1	msk-donskaya-sabdullakhi-sw-4	ICM
	0.002	msk-donskaya-sabdullakhi-sw-4	msk-donskaya-sabdullakhi-sw-1	ICM
	0.003	msk-donskaya-sabdullakhi-sw-1	msk-donskaya-sabdullakhi-gw-1	ICM
	0.004	msk-donskaya-sabdullakhi-gw-1	msk-donskaya-sabdullakhi-sw-1	ICM
	0.005	msk-donskaya-sabdullakhi-sw-1	msk-donskaya-sabdullakhi-sw-2	ICM
	0.006	msk-donskaya-sabdullakhi-sw-2	msk-donskaya-sabdullakhi-sw-3	ICM
	0.007	msk-donskaya-sabdullakhi-sw-3	sabdullakhi-mail	ICM
	0.008	sabdullakhi-mail	msk-donskaya-sabdullakhi-sw-3	ICM
	0.009	msk-donskaya-sabdullakhi-sw-3	msk-donskaya-sabdullakhi-sw-2	ICM
	0.010	msk-donskaya-sabdullakhi-sw-2	msk-donskaya-sabdullakhi-sw-1	ICM
	0.011	msk-donskaya-sabdullakhi-sw-1	msk-donskaya-sabdullakhi-gw-1	ICM
	0.012	msk-donskaya-sabdullakhi-gw-1	msk-donskaya-sabdullakhi-sw-1	ICM
	0.013	msk-donskaya-sabdullakhi-sw-1	msk-donskaya-sabdullakhi-sw-4	ICM
	0.014	msk-donskaya-sabdullakhi-sw-4	sabdullakhi-adm-donskaya-1	ICM
	0.878	--	msk-donskaya-sabdullakhi-sw-3	STP

Reset Simulation ☒ Constant Delay Captured t 0.878

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Анализ структуры передаваемого пакета

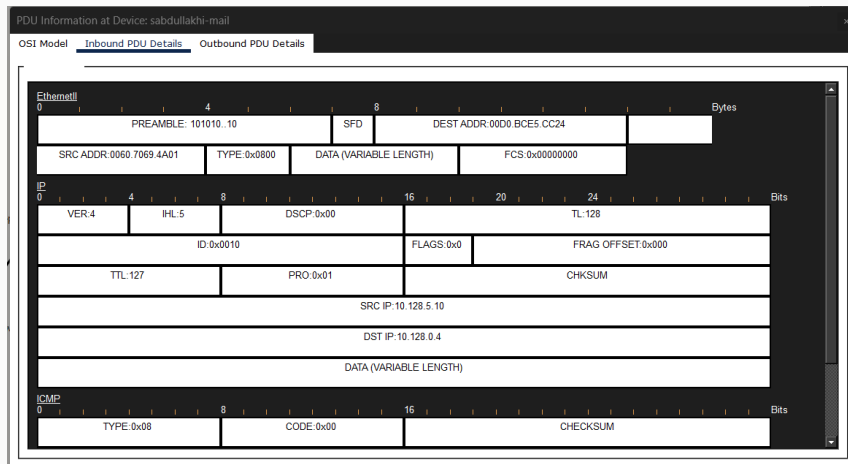


Рис. 9: Анализ структуры передаваемого ICMP пакета

В ходе лабораторной работы в Cisco Packet Tracer была построена сеть с маршрутизатором Cisco 2811, коммутаторами Cisco 2960, ПК и серверами. Маршрутизатор подключён к коммутатору через trunk-соединение для передачи трафика нескольких VLAN.

На маршрутизаторе выполнена базовая настройка (имя, пароли, SSH) и созданы виртуальные подинтерфейсы с IP-адресами для маршрутизации между VLAN.

Проверка командой ping подтвердила успешную передачу данных между VLAN. В режиме Simulation изучен процесс прохождения ICMP-пакета и структура кадров Ethernet, IP и ICMP.

Спасибо за внимание
